1. Air Cooling: Electricity Consumed

Adds up, month by month, the electricity used by the four parts of the cooling system: the air-handler fans, the chillers, the water pumps and the cooling-tower fans.

$$E_{cool} = \Sigma_m \{ e_{fan} \cdot Q_m + (Q_{chw,m} / 3.517) \times [f_{mech,m} \cdot k_{ch} + k_{pump} + k_{tower}] \}$$

where:

$e_{fan} = 3.412 \times SFP / (1.08 \times \Delta T_{air})$

$Q_{chw,m} = Q_m / \eta_{UPS} + e_{fan} \cdot Q_m$

$f_{mech,m} = 1 - \text{clamp}[(T_s + \Delta T_{chw} - T_{wb,m} - a_{tower} - a_{hx}) / \Delta T_{chw} , 0 , 1]$

$k_{pump} = (w_{chw} \times 24/\Delta T_{chw} + w_{cw} \times g_{cw}) / 1000$

$k_{tower} = 0.7457 \times g_{cw} / r_{tower}$

Parameters

Symbol	What it means (in plain words)	Value used
E_{cool}	Total electricity used for cooling over the whole period.	Result (kWh)
Σ_m	Add up the result for every month (m) of the year.	12 months
Q_m	Heat produced by the servers in month m. This comes from Module 1.	Input (kWh)
e_{fan}	Fan electricity needed for each unit of server heat.	0.197 / 0.118 / 0.079
SFP	How much electricity the air-handler fans use to move a given amount of air.	1.25 / 0.75 / 0.50 W/cfm
ΔT_{air}	How much the air warms up as it passes through the servers.	20 °F
$Q_{chw,m}$	Total heat the chilled water has to carry away: server heat, power-system losses and the heat from the fans themselves.	Calculated (kWh)
η_{UPS}	Efficiency of the backup power system (UPS). The small share of power it loses also turns into heat.	0.95
$f_{mech,m}$	Share of the month's cooling that needs the chiller. 0 means cool outdoor conditions do all the work for free; 1 means the chiller does it all.	0 to 1, by month
T_s	Temperature of the cold water sent to the air handlers.	55 °F
ΔT_{chw}	How much that cold water warms up before it returns to the plant.	10 °F
$T_{wb,m}$	Outdoor “wet-bulb” temperature for the month: one number that combines heat and humidity.	From weather data
a_{tower}	How close the cooling tower can bring its water to the wet-bulb temperature.	7 °F
a_{hx}	Small extra temperature gap lost in the heat exchanger between tower water and chilled water.	3 °F
$\text{clamp}[\dots , 0, 1]$	Keeps the result between 0 and 1.	—
k_{ch}	Electricity the chiller uses for each ton of cooling it delivers.	0.560 / 0.500 / 0.500 kW/ton
k_{pump}	Electricity all the water pumps use for each ton of cooling.	0.1098 kW/ton
w_{chw}, w_{cw}	Pump electricity per unit of water flow, for the chilled-water loop and the cooling-tower loop.	22 and 19 W/gpm
$24/\Delta T_{chw}$	Amount of chilled water that has to be pumped for each ton of cooling.	2.4 gpm/ton
g_{cw}	Amount of water pumped through the cooling tower for each ton of cooling.	3 gpm/ton
k_{tower}	Electricity the cooling-tower fans use for each ton of cooling.	0.0586 kW/ton
r_{tower}	Cooling-tower fan efficiency: how much water it cools per unit of fan power.	38.2 gpm/hp
$3.517, 3.412, 1.08, 0.7457$	Fixed conversion constants: 1 ton of cooling = 3.517 kW; kW to Btu/h; the heat-carrying constant of air; horsepower to kW.	Constants

2. Air Cooling: Water Consumed

Calculates the water lost in the cooling tower each month: the water that evaporates to carry the heat away, plus the tiny droplets blown out by the tower fans (drift).

$$W = \Sigma_m \{ 3600 \times Q_{rej,m} \times [f_{evap,m} / h_{fg}(T_{ao,m}) + d / (c_p \times \Delta T_{range})] \}$$

where:

$Q_{rej,m} = Q_{chw,m} + E_{ch,m} + E_{pump,m}$

$E_{ch,m} = (Q_{chw,m} / 3.517) \times f_{mech,m} \times k_{ch}$ $E_{pump,m} = (Q_{chw,m} / 3.517) \times k_{pump}$

$h_{fg}(T) = 2501 - 2.361 \times T$

$f_{evap,m} = h_{fg} \times (w_{out} - w_{in}) / (h_{a,out} - h_{a,in})$

$T_{ao,m} = T_{wb,m} + a_{tower} + \Delta T_{range} / 2$

Parameters

Symbol	What it means (in plain words)	Value used
W	Total water consumed (evaporation plus drift) over the period. 1 kg of water is about 1 litre.	Result (litres)
Σ_m	Add up the result for every month (m) of the year.	12 months
$Q_{rej,m}$	Heat that reaches the cooling tower in month m. This is what links the water result to the electricity result.	Calculated (kWh)
$Q_{chw,m}$, $E_{ch,m}$, $E_{pump,m}$	Chilled-water heat, chiller electricity and pump electricity from the electricity equation. The chiller and pump energy also ends up as heat in the tower.	From Section 1 (kWh)
3600	Converts kWh into kJ (1 kWh = 3,600 kJ).	Constant
$f_{evap,m}$	Share of the tower's heat that is removed by evaporating water. The rest simply warms the outdoor air and uses no water.	0.61 to 0.85, by month
h_{fg}	Energy needed to turn 1 kg of water into vapour at cooling-tower temperature.	2,437 to 2,495 kJ/kg
$2501, 2.361$	Fitted constants that give h_{fg} at any temperature (matches steam tables).	Constants
$T_{ao,m}$	Temperature of the air leaving the tower, taken as the average water temperature inside the tower.	Calculated (°C)
w_{in}, w_{out}	Amount of moisture carried by the air entering and leaving the tower.	Standard moist-air formulas
$h_{a,in}$, $h_{a,out}$	Total heat content of the air entering and leaving the tower.	Standard moist-air formulas
$T_{db,m}$, RH_m	Monthly outdoor air temperature and relative humidity, used to work out w_{in} and $h_{a,in}$.	From weather data
p	Air pressure, used in the moist-air formulas.	101.325 kPa
$T_{wb,m}$	Monthly outdoor wet-bulb temperature (heat and humidity combined).	From weather data
a_{tower}	How close the tower can cool its water to the wet-bulb temperature.	7 °F (3.9 °C)
d	Drift: share of the circulating water blown out of the tower as droplets.	0.0005%
c_p	Energy needed to warm 1 kg of water by 1 °C.	4.186 kJ/kg·°C
ΔT_{range}	How much the tower cools the water passing through it.	10 °F (5.6 °C)

3. Direct-to-Chip Cooling: Electricity Consumed

Liquid flowing through metal plates fixed on the chips captures most of the heat; a small air-cooling system removes the rest. The equation adds up six electricity users for every time step.

$$E_{cool} = \Sigma_t \{ E_{CDU,t} + E_{FWS,t} + E_{dry,t} + E_{ch,liq,t} + E_{twr,t} + E_{air,t} \}$$

where:

$Q_{liq,t} = f_{cap} \times Q_{IT,t}$ $Q_{air,t} = (1 - f_{cap}) \times Q_{IT,t}$ $Q_{UPS,t} = Q_{IT,t} \times (1/\eta_{UPS} - 1)$

$E_{CDU,t} = e_{CDU} \times Q_{liq,t}$ $E_{FWS,t} = e_{FWS} \times Q_{liq,t}$ $Q'_{liq,t} = Q_{liq,t} + E_{CDU,t} + E_{FWS,t}$

$T_{ret} = T_{fws} + \Delta T_{FWS}$ $T_1 = \text{clamp}(T_{db,t} + a_{dry} , T_{fws} , T_{ret})$ $T_2 = \text{clamp}(T_{wb,t} + a_{tower} + a_{hx} , T_{fws} , T_1)$

$f_{dry} = (T_{ret} - T_1) / \Delta T_{FWS}$ $f_{twr} = (T_1 - T_2) / \Delta T_{FWS}$ $f_{ch} = (T_2 - T_{fws}) / \Delta T_{FWS}$

$E_{dry,t} = f_{dry} \times Q'_{liq,t} / R$ $E_{ch,liq,t} = (f_{ch} \times Q'_{liq,t} / 3.517) \times k_{ch}$

$E_{twr,t} = (Q_{twr,t} / 3.517) \times (k_{tower} + k_{cw})$, with $Q_{twr,t} = (f_{twr} + f_{ch}) \times Q'_{liq,t} + E_{ch,liq,t}$

$E_{air,t} = e_{fan} \cdot Q_{air,t} + (Q_{chw,air,t} / 3.517) \times [f_{mech,air,t} \cdot k_{ch} + k_{pump} + k_{tower}]$, with $Q_{chw,air,t} = Q_{air,t} + Q_{UPS,t} + e_{fan} \cdot Q_{air,t}$

e_{fan}, $f_{mech,air,t}$ and k_{pump} use the same formulas as in Section 1.

Parameters

Symbol	What it means (in plain words)	Value used
E_{cool}	Total electricity used for cooling over the whole period.	Result (kWh)
Σ_t	Add up for every time step t (an hour, or a group of similar-weather hours).	Each time step
$E_{CDU,t}$	Pumps in the coolant unit that push coolant through the chip cold plates.	Calculated (kWh)
$E_{FWS,t}$	Pumps that move the building's water loop to the outdoor equipment.	Calculated (kWh)
$E_{dry,t}$	Dry-cooler fans (outdoor radiators that cool water with air, using no water).	Calculated (kWh)
$E_{ch,liq,t}$	Chiller on the liquid side, used only when the weather is very hot.	Calculated (kWh)
$E_{twr,t}$	Cooling-tower fans and pumps on the liquid side.	Calculated (kWh)
$E_{air,t}$	The small air-cooling system that removes the heat the cold plates miss.	Calculated (kWh)
$Q_{IT,t}$	Heat produced by the servers in time step t (from Module 1).	Input (kWh)
f_{cap}	Share of server heat captured by the cold plates; the rest goes to room air.	0.70 / 0.80 / 0.94
$Q_{liq,t}$, $Q_{air,t}$	Heat that goes into the liquid, and heat that goes into the room air.	Calculated (kWh)
$Q_{UPS,t}$, η_{UPS}	Heat lost in the backup power system (UPS), and that system's efficiency.	η_{UPS} = 0.95
e_{CDU}	Coolant-unit pump electricity per unit of liquid heat (from a CDU rating).	0.00612 kWh/kWh
e_{FWS}	Building-water pump electricity per unit of liquid heat.	0.00527 kWh/kWh
$Q'_{liq,t}$	Liquid heat plus the heat added by the pumps: the total to be sent outdoors.	Calculated (kWh)
T_{fws}	Temperature of building water sent to the coolant units (warmer is easier).	27 / 32 / 40 °C
ΔT_{FWS}, T_{ret}	How much the building water warms up, and the temperature it returns at.	14.1 °C; calculated
$T_{db,t}$, $T_{wb,t}$	Outdoor air temperature, and wet-bulb temperature (heat and humidity combined).	From weather data
a_{dry}	How close the dry cooler can bring the water to the outdoor air temperature.	10 °C
a_{tower}, a_{hx}	How close the tower gets to wet-bulb, plus the gap lost in its heat exchanger.	7 °F, 3 °F
T_1, T_2	Water temperature after the dry cooler, and after the cooling tower.	Calculated (°C)
f_{dry}, f_{twr}, f_{ch}	Shares of liquid heat removed by dry cooler, tower and chiller (sum to 1).	Varies with weather
clamp( )	Keeps a temperature between a lower and an upper limit.	—
R	Dry-cooler efficiency: heat removed per unit of fan electricity.	45 / 70 / 110
k_{ch}	Electricity the chiller uses for each ton of cooling.	0.560 / 0.500 / 0.500 kW/ton
$Q_{twr,t}$	Liquid-side heat sent to the cooling tower, including the chiller's own heat.	Calculated (kWh)
k_{tower}, k_{cw}	Electricity per ton of cooling used by the tower fans, and by the tower pumps.	0.0556, 0.057 kW/ton
$Q_{chw,air,t}$	Heat the air system must remove: leftover server heat, UPS losses, fan heat.	Calculated (kWh)
e_{fan}, $f_{mech,air,t}$, k_{pump}	Air-side fan electricity per unit of heat, share needing the chiller, and pump electricity per ton (as in Section 1).	As in Section 1
3.517	Converts kWh of heat into ton-hours (1 ton of cooling = 3.517 kW).	Constant

4. Direct-to-Chip Cooling: Water Consumed

The coolant loop, the building water loop and the dry coolers are closed and use no water. Water is consumed only in the cooling towers, which serve the liquid side (on warm hours) and the small air-cooling system.

$$W = \Sigma_t \{ 3600 \times Q_{twr,t} \times [f_{evap,liq,t} / h_{fg}(T_{ao,liq}) + d / (c_p \times \Delta T_{range})] + 3600 \times Q_{rej,air,t} \times [f_{evap,air,t} / h_{fg}(T_{ao,air}) + d / (c_p \times \Delta T_{range})] \}$$

where:

$Q_{rej,air,t} = Q_{chw,air,t} + E_{ch,air,t} + E_{pump,air,t}$

$E_{ch,air,t} = (Q_{chw,air,t} / 3.517) \times f_{mech,air,t} \times k_{ch}$ $E_{pump,air,t} = (Q_{chw,air,t} / 3.517) \times k_{pump}$

$T_{ao,liq} = (T_2 - a_{hx}) + \Delta T_{range} / 2$ $T_{ao,air} = T_{wb,t} + a_{tower} + \Delta T_{range} / 2$

$h_{fg}(T) = 2501 - 2.361 \times T$

$f_{evap} = h_{fg}(T_{ao}) \times (w_{out} - w_{in}) / (h_{a,out} - h_{a,in})$

Parameters

Symbol	What it means (in plain words)	Value used
W	Total water consumed by the cooling towers (evaporation plus drift). 1 kg of water is about 1 litre.	Result (litres)
Σ_t	Add up the result for every time step t.	Each time step
$Q_{twr,t}$	Liquid-side heat sent to the cooling tower. It is zero whenever the dry cooler can handle all the heat.	From Section 3 (kWh)
$Q_{rej,air,t}$	Heat from the small air-cooling system that reaches its cooling tower.	Calculated (kWh)
$Q_{chw,air,t}$, $E_{ch,air,t}$, $E_{pump,air,t}$	The air system's chilled-water heat, chiller electricity and pump electricity. The chiller and pump energy also ends up as heat in the tower.	From Section 3 (kWh)
3600	Converts kWh into kJ (1 kWh = 3,600 kJ).	Constant
$f_{evap,liq,t}$, $f_{evap,air,t}$	Share of each tower's heat that is removed by evaporating water. The rest simply warms the outdoor air.	Calculated each time step
h_{fg}	Energy needed to turn 1 kg of water into vapour at cooling-tower temperature.	About 2,430 kJ/kg at 30 °C
$2501, 2.361$	Fitted constants that give h_{fg} at any temperature (matches steam tables).	Constants
$T_{ao,liq}$, $T_{ao,air}$	Temperature of the air leaving each tower, taken as the average water temperature inside it.	Calculated (°C)
T_2	Building-water temperature after the cooling-tower stage.	From Section 3 (°C)
a_{tower}, a_{hx}	How close the tower gets to the wet-bulb temperature, plus the small gap lost in its heat exchanger.	7 °F, 3 °F (3.9 °C, 1.7 °C)
$T_{wb,t}$	Outdoor wet-bulb temperature (heat and humidity combined).	From weather data
$T_{db,t}$	Outdoor air temperature, used with humidity to work out w_{in} and $h_{a,in}$.	From weather data
w_{in}, w_{out}	Amount of moisture carried by the air entering and leaving the tower.	Standard moist-air formulas
$h_{a,in}$, $h_{a,out}$	Total heat content of the air entering and leaving the tower.	Standard moist-air formulas
p	Average air pressure at the site, used in the moist-air formulas.	100.5 kPa
d	Drift: share of the circulating water blown out of the tower as droplets.	0.0005%
c_p	Energy needed to warm 1 kg of water by 1 °C.	4.186 kJ/kg·°C
ΔT_{range}	How much the tower cools the water passing through it.	10 °F (5.6 °C)